

IEEE 829 Test Plan

Lung Cancer Risk Prediction System

Shrinidhi R Raikar
Shashank Gautham
Nithish Nagendra
Pruthviraj R

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1 Test Plan Identifier

- **Test Plan ID:** TP-LCRP-001
- **Project:** Lung Cancer Risk Prediction System
- **Version:** 1.0

2 References

- Project Requirements Document
- System Design Document
- Dataset Documentation
- IEEE 829 Software Testing Standard

3 Introduction

This document defines the testing strategy for the Lung Cancer Risk Prediction System developed using Machine Learning and Deep Learning techniques. The goal is to ensure the system functions correctly, provides accurate predictions, and meets performance and security requirements.

4 Test Items

The following modules are tested:

Web Application User Interface

- Input Validation Module
- Flask Backend API
- Machine Learning Prediction Models
- Deep Learning Model
- Dataset Processing
- Result Display Module

5 Software Risk Issues

Possible risks during testing include:

- Incorrect prediction results
- Data processing errors
- Server crashes
- Security vulnerabilities (XSS, SQL Injection)
- Browser compatibility issues

6 Features to be Tested

- Homepage loading
- User login/signup
- Input validation
- Prediction functionality (ML, DL, QML)
- Result display (prediction + confidence score)
- System performance
- Security protection

7 Features not to be Tested

- Internal working of third-party libraries (TensorFlow, Scikit-learn)
- Hardware-level performance
- External network infrastructure

8 Approach

Testing will be carried out using:

- Functional Testing
- Integration Testing
- System Testing
- Performance Testing
- Security Testing

Test cases will be executed manually and verified against expected results.

9 Item Pass/Fail Criteria

A test case is considered PASS if the actual output matches the expected output. If the expected output is not obtained, the test case will be marked as FAIL.

10 Suspension Criteria and Resumption Requirements

Testing will be suspended if major issues such as server crash, model loading failure, or database error occur. Testing will resume once the issue is fixed.

11 Test Deliverables

- Test Plan Document
- Test Case Specification
- Test Execution Report
- Bug Reports
- Final Test Summary

12 Remaining Test Tasks

Remaining tasks include regression testing, system performance testing and final validation before deployment.

13 Environmental Needs

Hardware Requirements

- Minimum 8GB RAM

- Intel i5 Processor or higher

Software Requirements

- Python
- Flask Framework
- Web Browser (Google Chrome)
- Machine Learning Libraries (Scikit-learn, TensorFlow)

14 Staffing and Training Needs

Testing is performed by the development team. Basic knowledge of Python, Flask, and Machine Learning is required.

15 Responsibilities

- Developer – System implementation and bug fixing
- Tester – Execution of test cases
- Project Guide – Review and approval of testing process

16 Schedule

Task	Duration
Test Plan Preparation	2 Days
Test Case Design	3 Days
Test Execution	4 Days
Bug Fixing	2 Days
Final Testing	2 Days

17 Planning Risks and Contingencies

Possible risks include dataset inconsistencies, prediction errors, and server downtime. Backup datasets and error handling mechanisms are used to reduce these risks.

18 Test Case Specification

TC ID	Description	Expected Result	Actual Result	Status
TC01	Homepage loads	Opens successfully	Opened	Pass
TC02	Input form fields	All fields visible	Displayed	Pass
TC03	Empty input validation	Error message shown	Shown	Pass
TC04	Valid input	Accepted	Accepted	Pass
TC05	Invalid input	Error displayed	Shown	Pass
TC06	Prediction output	Result displayed	Displayed	Pass
TC07	Confidence score	Percentage shown	Shown	Pass
TC08	SQL Injection	Prevented	Prevented	Pass

TC ID	Description	Expected Result	Actual Result	Status
TC09	XSS attack	Blocked	Blocked	Pass
TC10	End-to-end workflow	Works correctly	Successful	Pass

19 Test Execution Summary

Total Test Cases Designed	30
Test Cases Executed	30
Test Cases Passed	30
Test Cases Failed	0
Success Rate	100%

20 Conclusion

All test cases were executed successfully and the system produced the expected results. The Lung Cancer Risk Prediction System is reliable, secure and ready for deployment.

21 Approvals

Name	Role
Shrinidhi R Raikar	Developer
Project Guide	Supervisor